

CAT

Advanced Computer Architecture

Homework #1 Multithreading (25%)

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Due on 13 May 2026 @14:00

1 Introduction

In Lab #2, you got hands-on with the AlphaBot2's functionalities and attempted to navigate it along a black track on the floor.

2 Tasks

Use multithreading to implement the tasks of AlphaBot2:

1. It should continuously follow the black track (no veering off the track).
2. It detects using the infrared sensors. When it detects the 1st object, it buzzes once. When it detects the 2nd object, it buzzes twice. When it detects the 3rd object, it buzzes thrice.
3. It recognizes objects placed in the line of sight of the camera (not on the line). The three objects are a shoe, a bottle, and a mug. When it recognizes a shoe, it sets the RGB LED red. When it recognizes the bottle, it sets the RGB LED yellow. When it recognizes the mug, it sets the RGB LED green.

You can place the objects at convenient spots to facilitate recognition. Please time AlphaBot2 for completing a round trip on the track. You can either use the timer on your mobile or use the timing built-in functions in Python.

Note: You can run the calibration once and save the minimum and maximum sensor values to be used for all subsequent runs.

3 Deliverables

Submission requirements: The report must not exceed 5 pages. Include your names, matriculation ID, group number, and robot number.

3.1 Report

- **Design** Implementation of Multithreading or Multiprocessing in Python (including the distribution of tasks).
- **Results** Analysis of the tasks' performance compared to the sequential version (how performance improved and why).

Please submit your program as well.

3.2 Demo during lab

- **Demo #1:** AlphaBot2 can follow the line.
- **Demo #2:** AlphaBot2 can detect objects and recognize them.
- **Demo #3:** AlphaBot2 buzzes and set the LEDs correctly.

Grading Scheme for Homework #1

Functionality	Marks
Follow track with at most one slight deviation	20
Object recognition (3 objects)	30
Buzz correct number of times (3 objects)	10
Display RGB LED color (3 objects)	10
Report	30
Total	100

4 Optional: Robot Race (27 May 2026 @13:30)

The track will be more complex, including branches. Before the race, we will post the details on Moodle. Please check out. The group whose AlphaBot2 successfully completes the path with the lowest time will be the winner!

Bonus 5 points for winners!